Computer Vision – Lecture 3

Image Processing II

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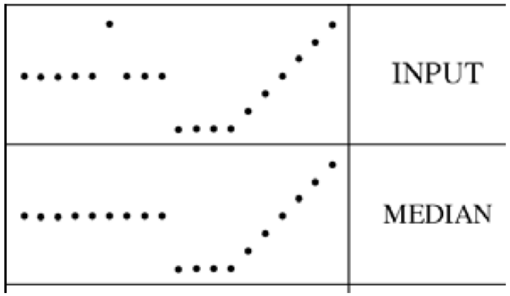
Computer Vision

Course Outline

- Image Processing Basics
  - Image Formation
  - Linear Filters
  - Gradient Filters and Edge Detection
  - Structure Extraction
- Segmentation
- Object Recognition and Categorization
- Deep Learning Approaches for Vision
- 3D Reconstruction

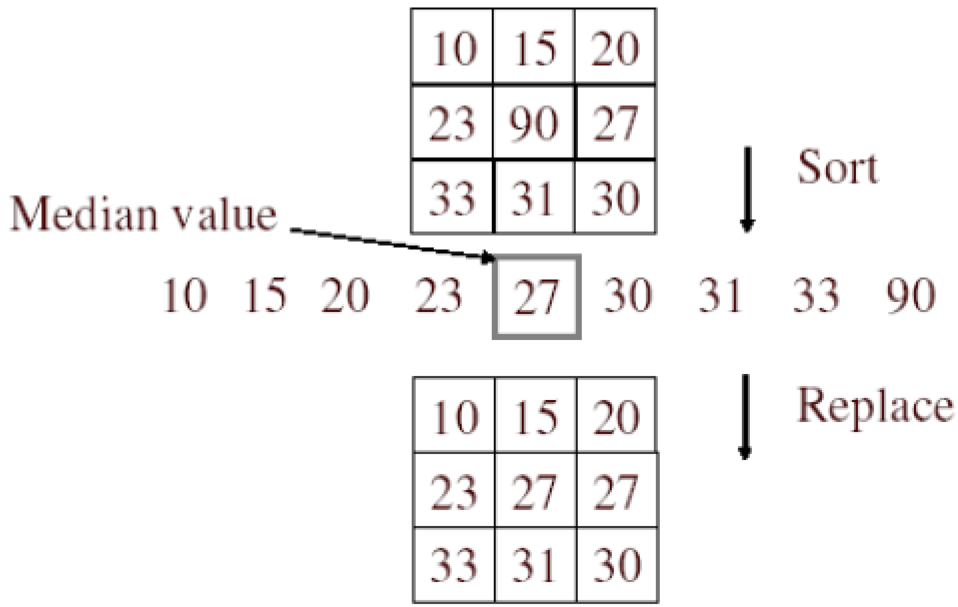
Topics of This Lecture

- Non-linear Filters
  - Median Filter
- Multi-Scale representations
  - How to properly rescale an image?
- Filters as templates
  - Correlation as template matching
- Image gradients
  - Derivatives of Gaussian



Non-Linear Filters: Median Filter

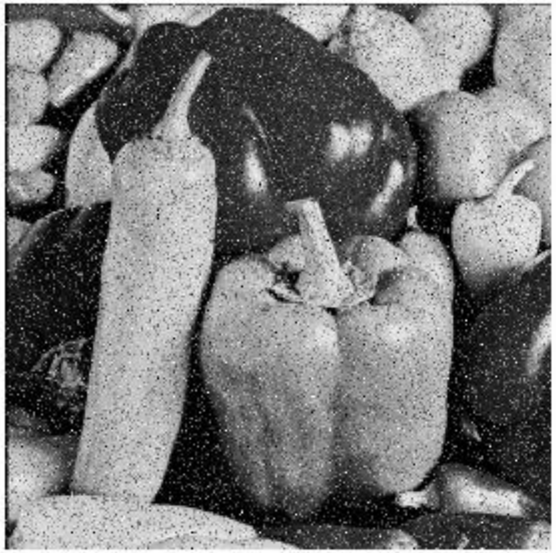
- Basic idea
  - Replace each pixel by the median of its neighbors.
- Properties
  - Doesn't introduce new pixel values
  - Removes spikes: good for impulse, salt & pepper noise
  - Linear?

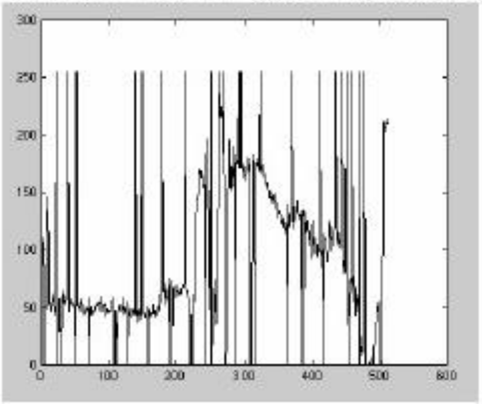


Slide credit: Kristen Grauman

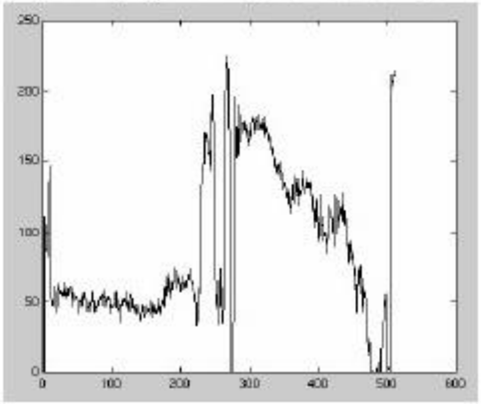
Median Filter

Salt and pepper noise





Median filtered



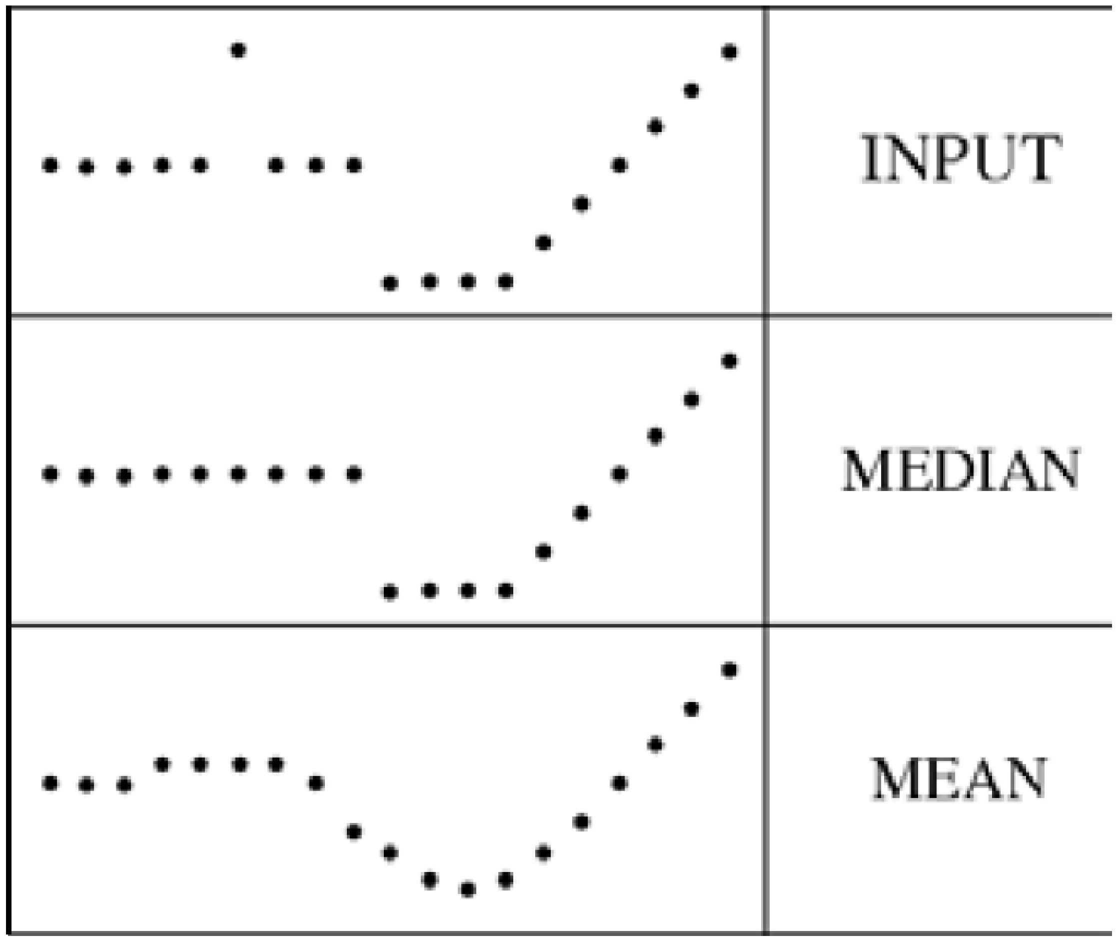
Plots of a row of the image

Slide credit: Kristen Grauman

Image Source: Martial Hebert

Median Filter

- The Median filter is edge preserving.



Slide credit: Kristen Grauman

Median vs. Gaussian Filtering

3x3 | 5x5 | 7x7

Gaussian

Median

Slide credit: Svetlana Lazebnik

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Motivation: Fast Search Across Scales

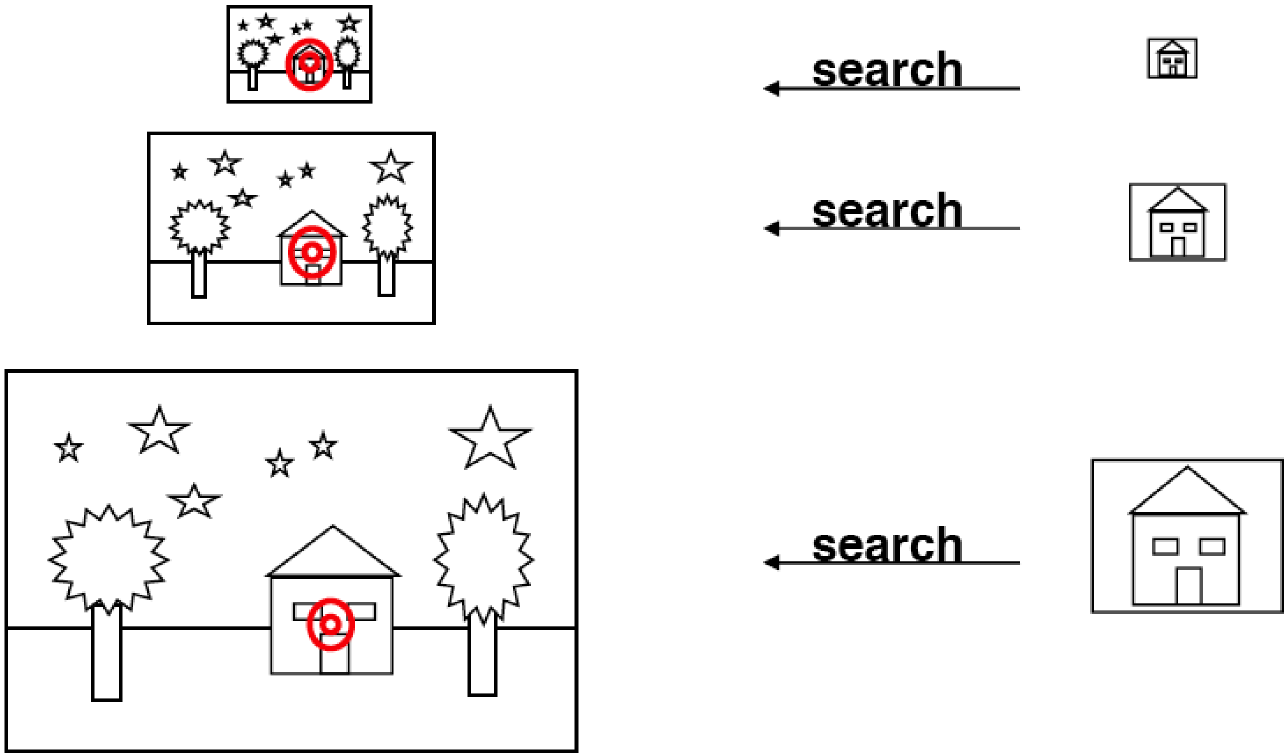


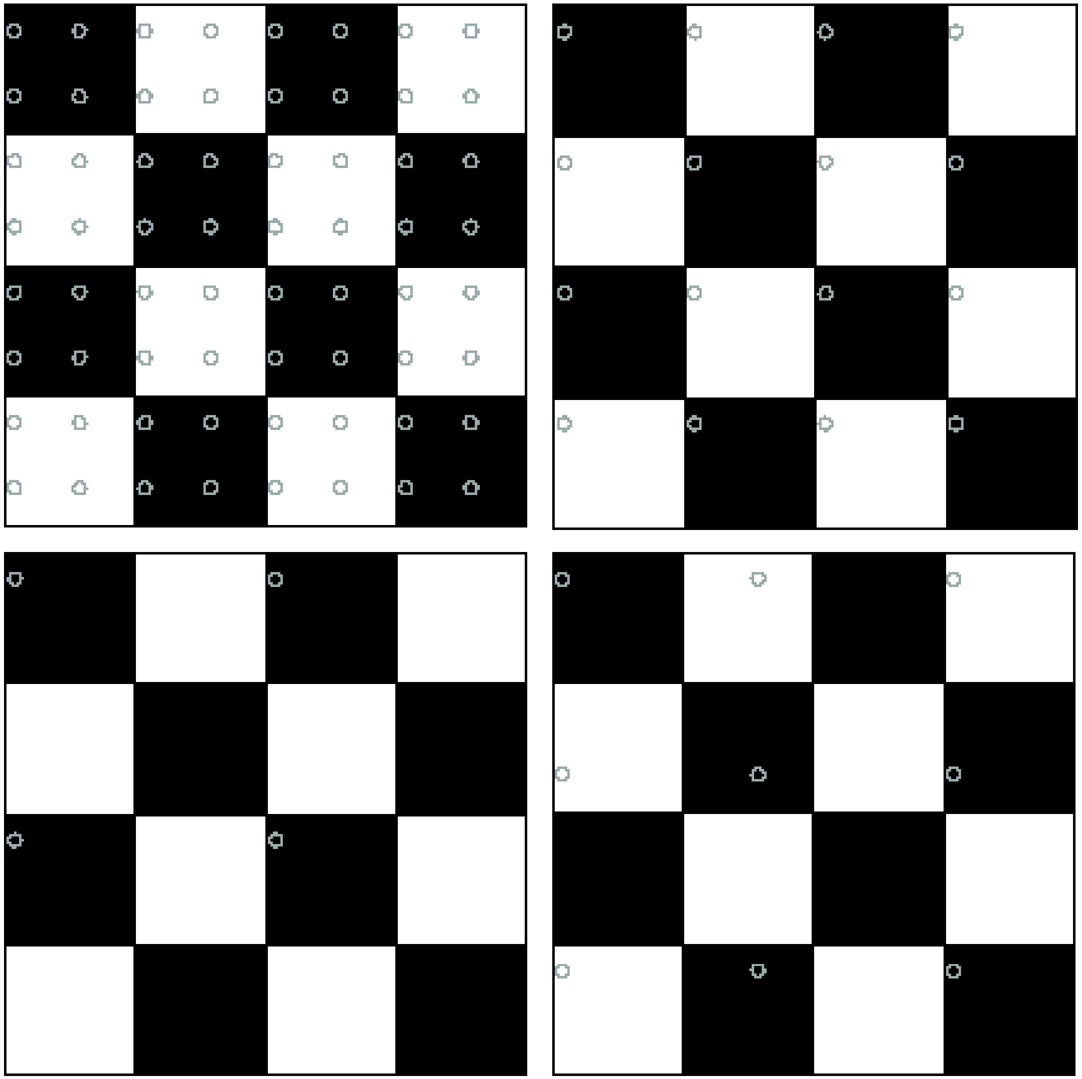
Image Source: Irani & Basri

Image Pyramid

Low resolution

High resolution

How Should We Go About Resampling?



Let's resample the checkerboard by taking one sample at each circle.

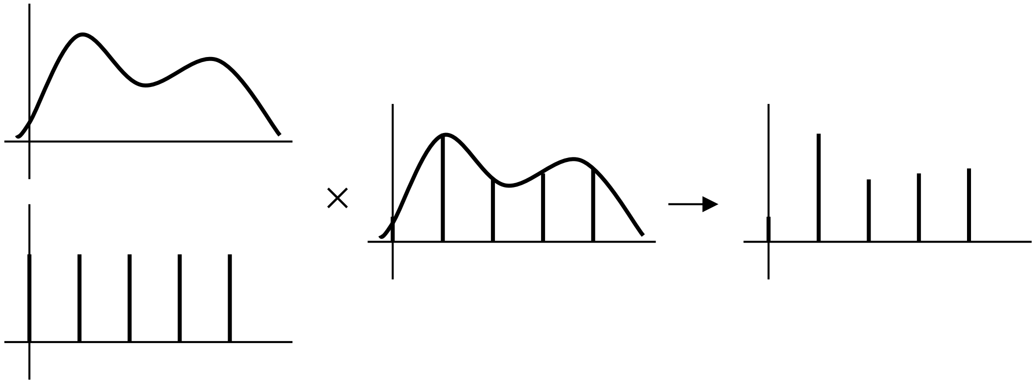
In the top left board, the new representation is reasonable. Top right also yields a reasonable representation.

Bottom left is all black (dubious) and bottom right has checks that are too big.

Image Source: Forsyth & Ponce

Fourier Interpretation: Discrete Sampling

- Sampling in the spatial domain is like multiplying with a spike function.



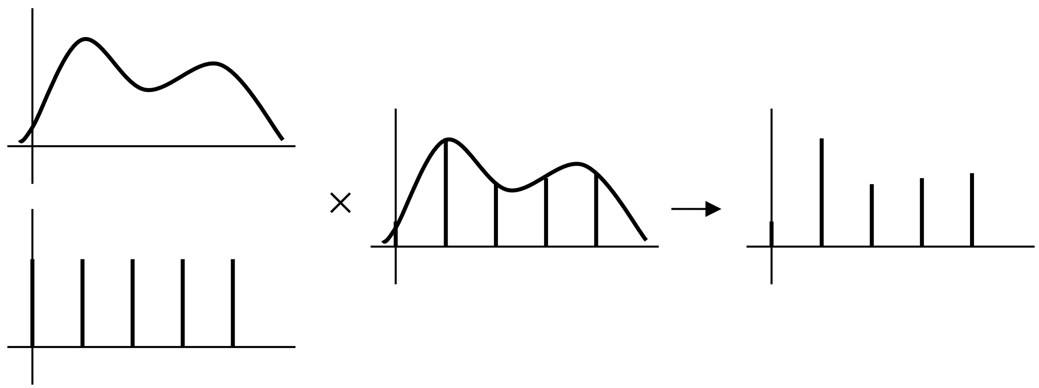
- Sampling in the frequency domain is like...

?

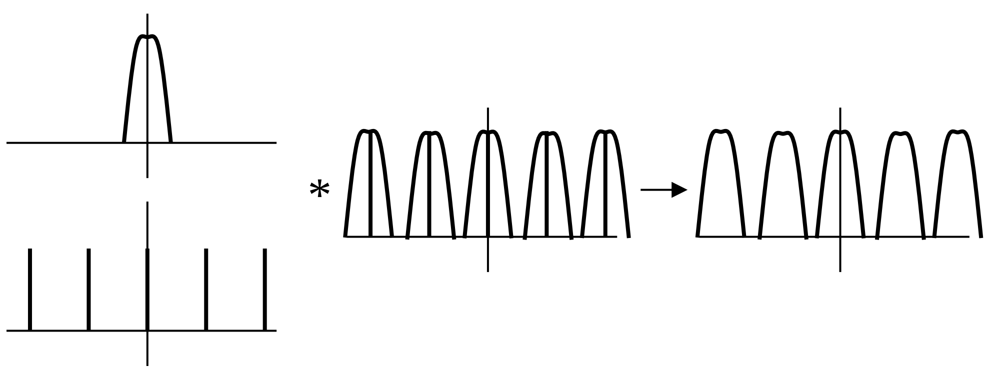
Source: S. Chenney

Fourier Interpretation: Discrete Sampling

- Sampling in the spatial domain is like multiplying with a spike function.



- Sampling in the frequency domain is like convolving with a spike function.



Source: S. Chenney

Sampling and Aliasing

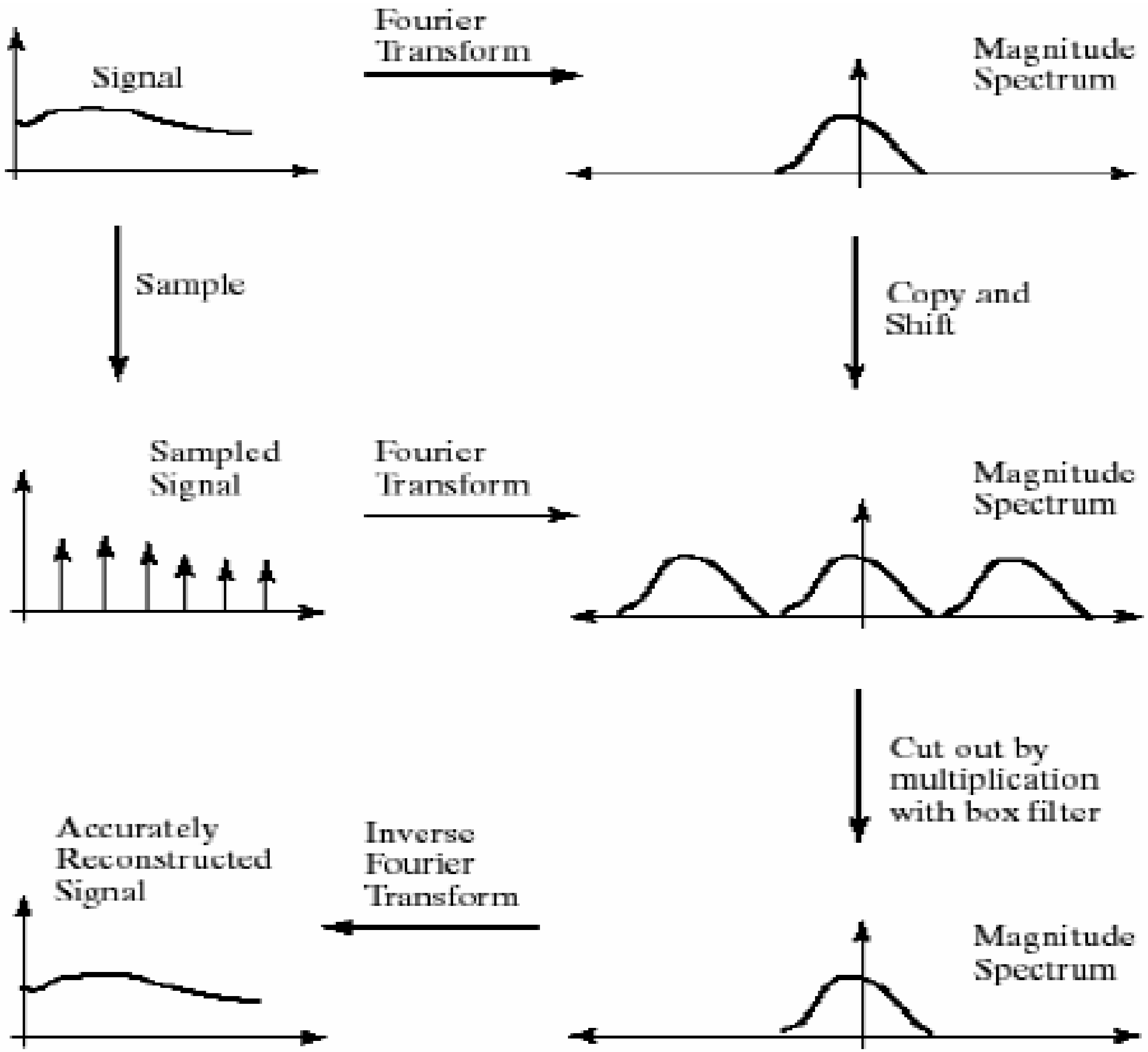
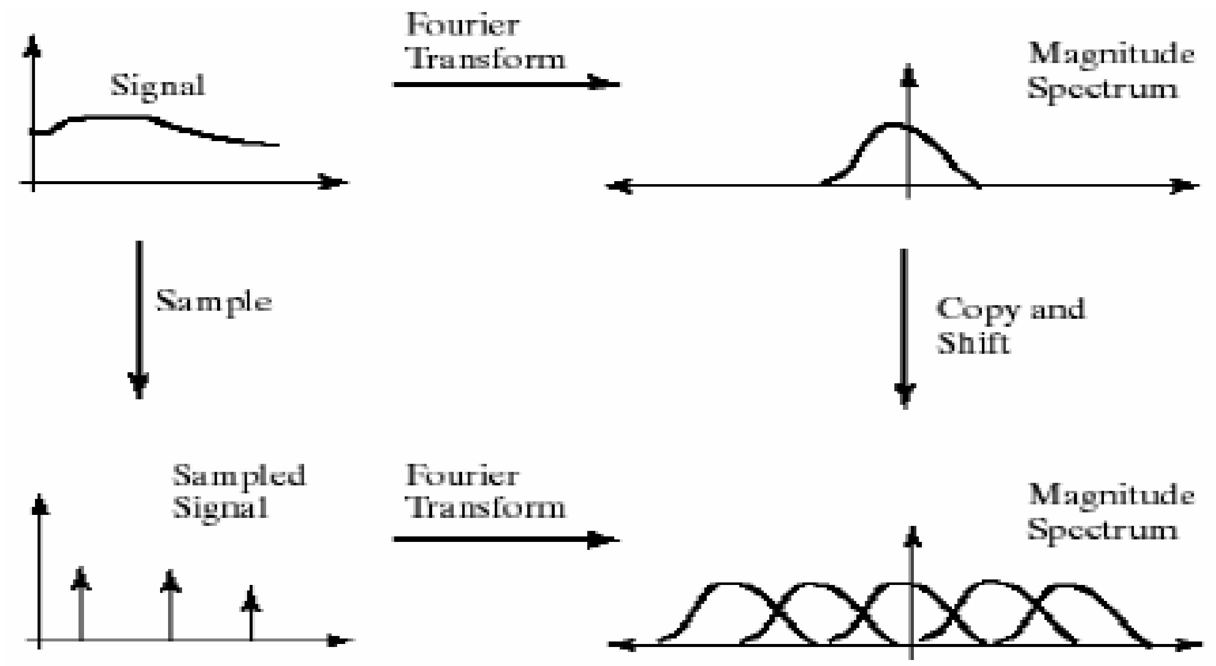


Image Source: Forsyth & Ponce

Sampling and Aliasing



- Nyquist theorem:
  - In order to recover a certain frequency f, we need to sample with at least $2f$.
  - This corresponds to the point at which the transformed frequency spectra start to overlap (the Nyquist limit)

Image Source: Forsyth & Ponce

Sampling and Aliasing

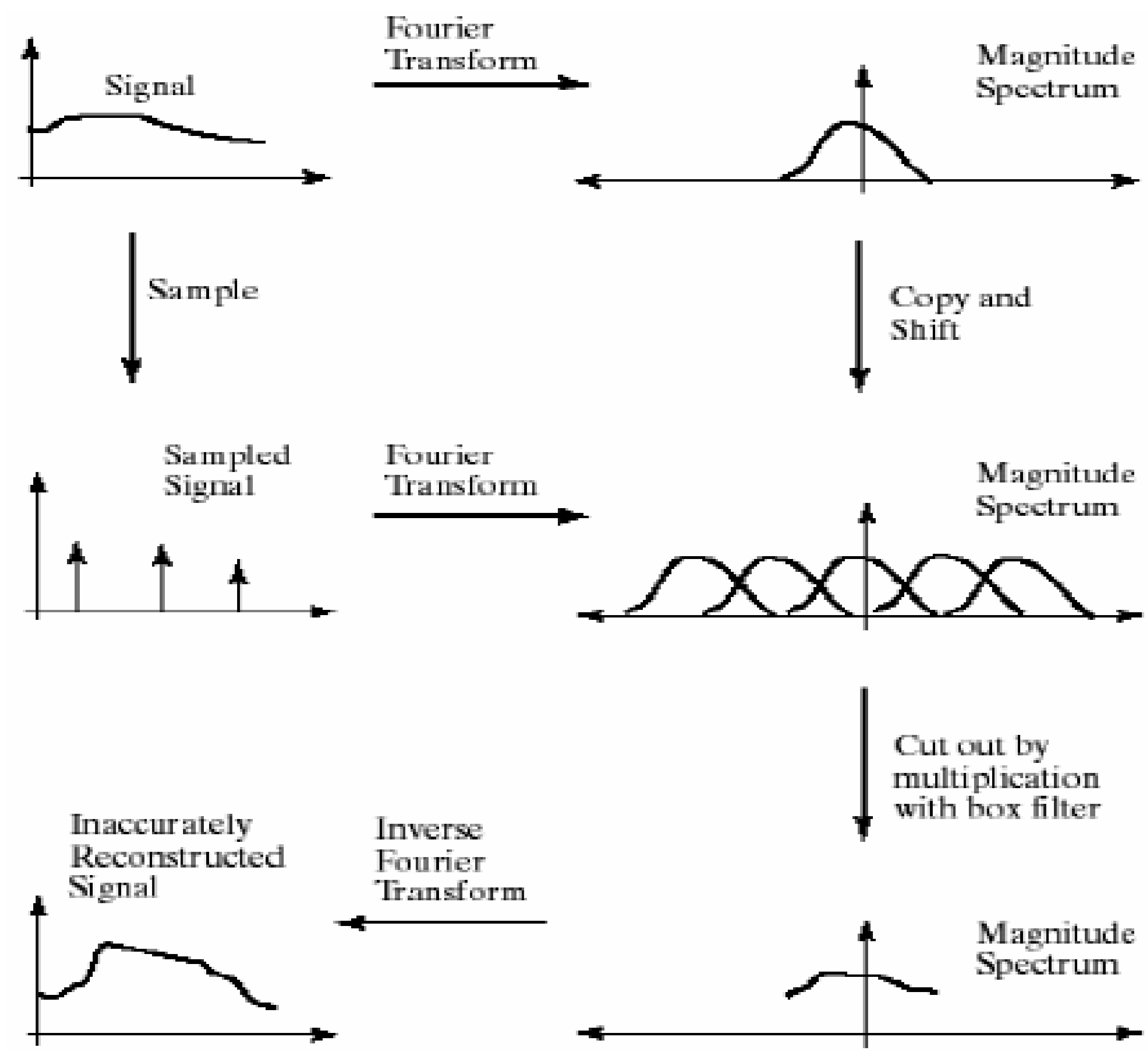
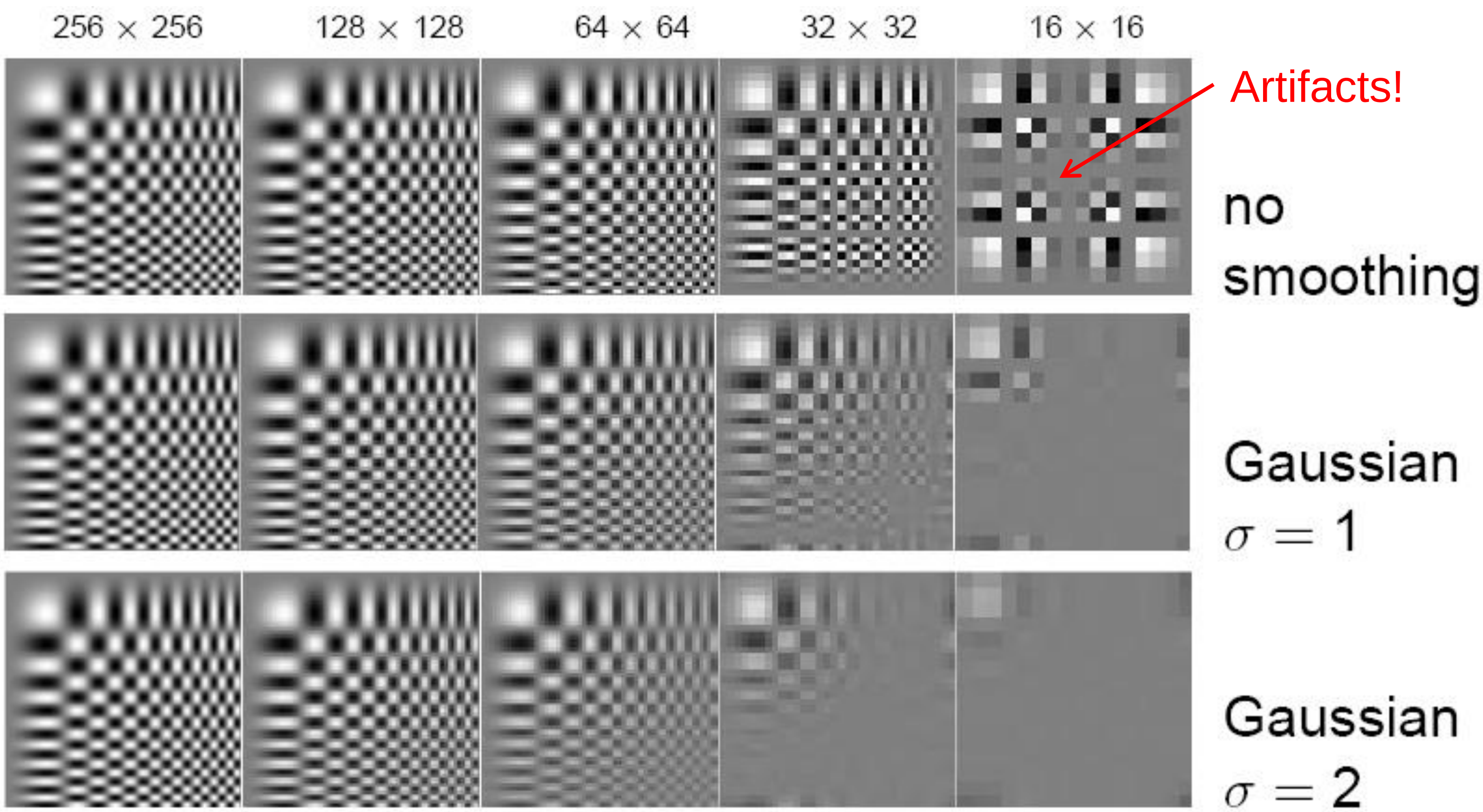


Image Source: Forsyth & Ponce

Aliasing in Graphics

Image Source: Alexej Efros

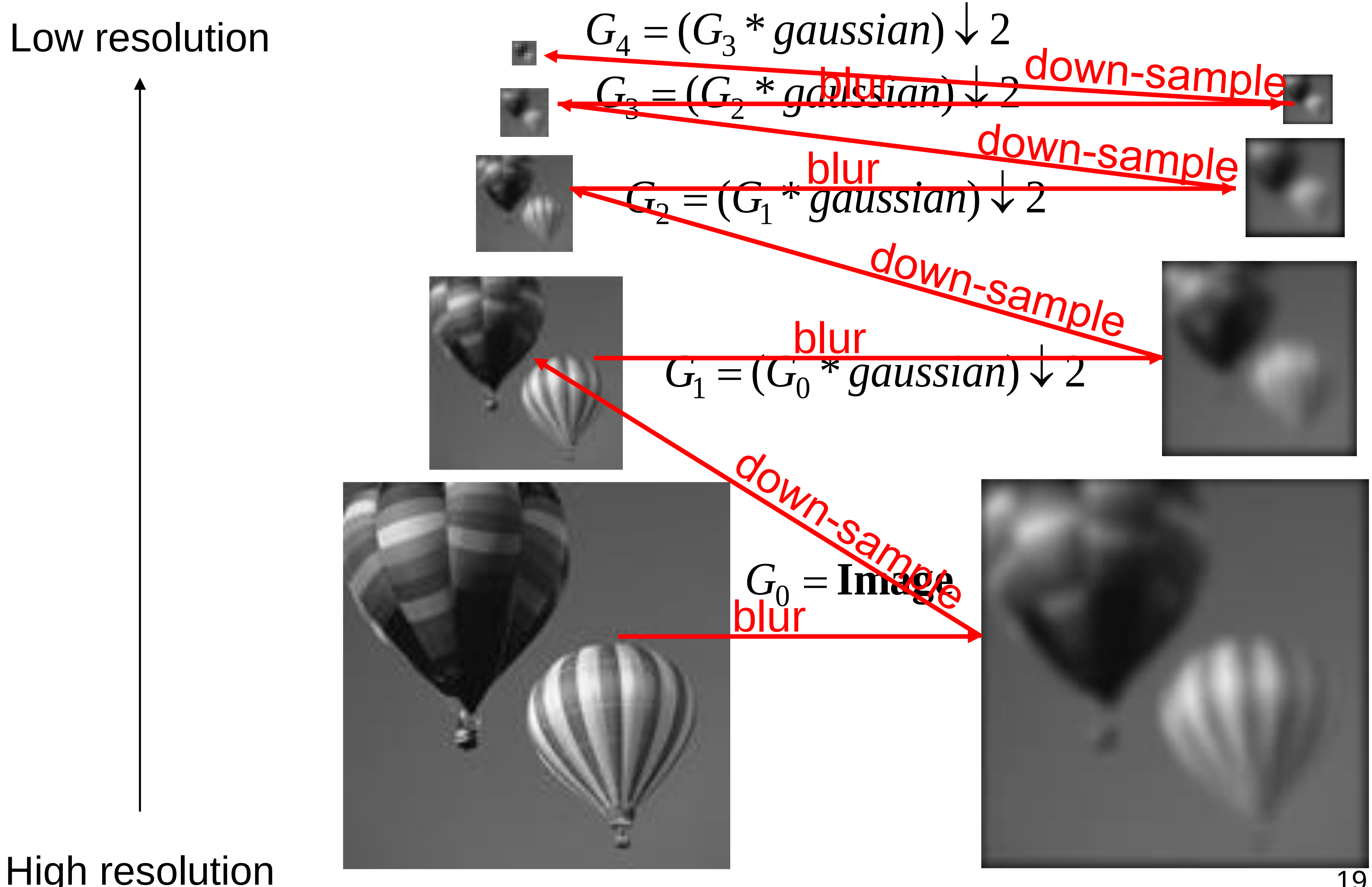
Resampling with Prior Smoothing



- Note: We cannot recover the high frequencies, but we can avoid artifacts by smoothing before resampling.

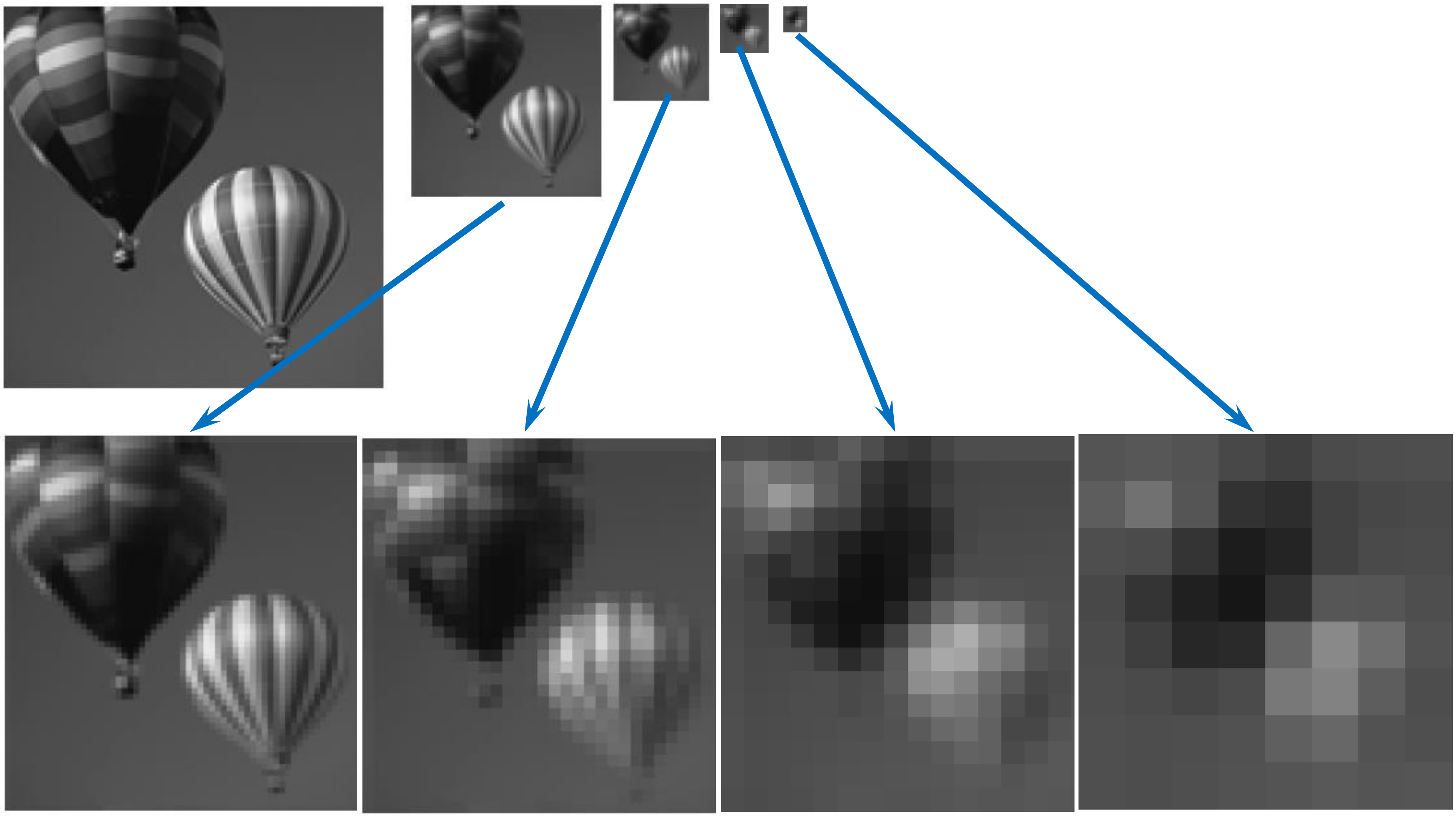
Image Source: Forsyth & Ponce

The Gaussian Pyramid



Source: Irani & Basri

Gaussian Pyramid – Stored Information



Source: Irani & Basri

Summary: Gaussian Pyramid

- Construction: create each level from previous one
  - Smooth and sample
- Smooth with Gaussians, in part because
  - a Gaussian*Gaussian = another Gaussian
  - $G(\sigma_1) * G(\sigma_2) = G(\mathrm{sqrt}(\sigma_1^{2} + \sigma_2^{2}))$
- Gaussians are low-pass filters, so the representation is redundant once smoothing has been performed.

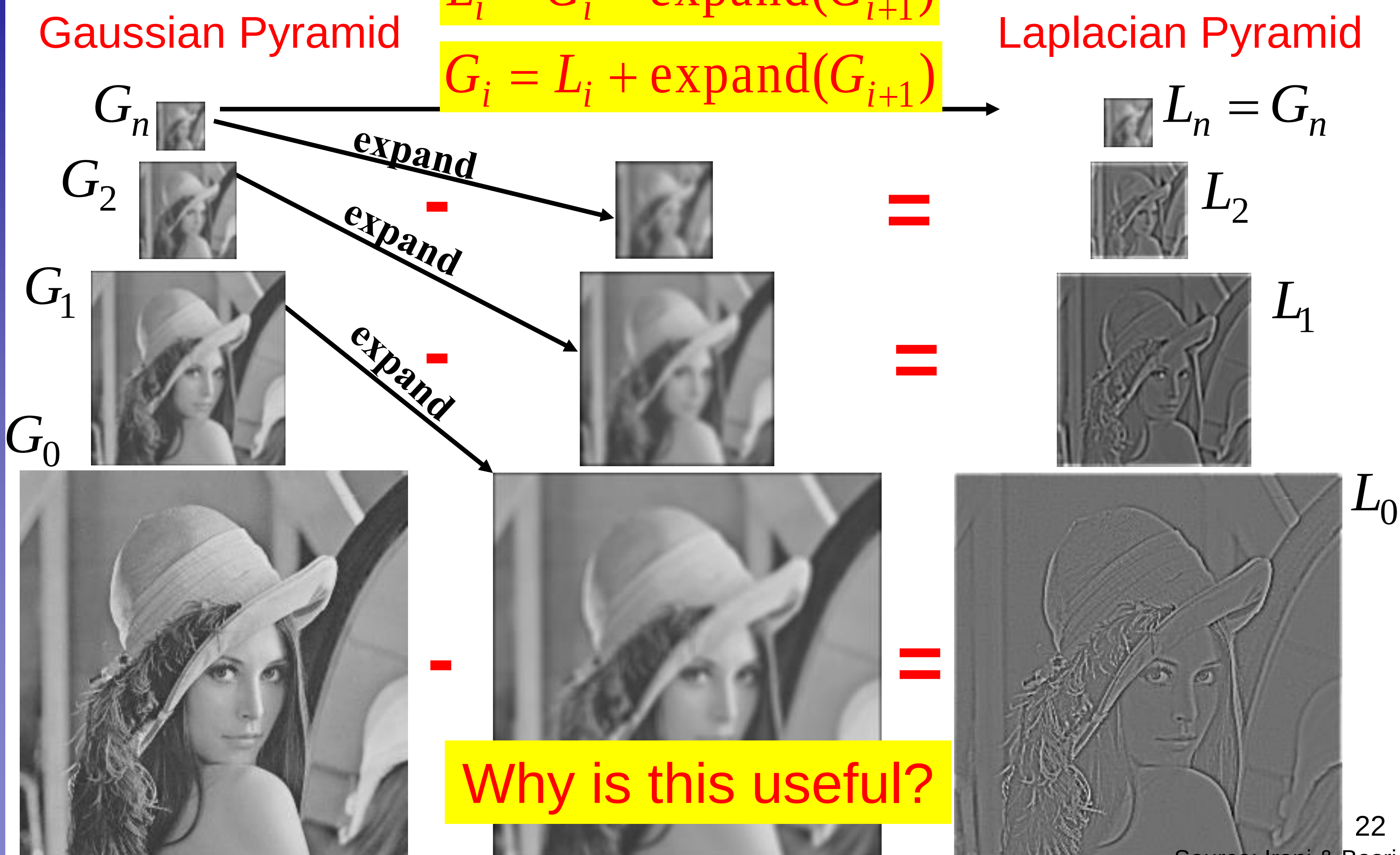
  $\Rightarrow$ There is no need to store smoothed images at the full original resolution.

Slide credit: David Lowe

The Laplacian Pyramid

$$L_i = G_i - \mathrm{expand}(G_{i+1})$$

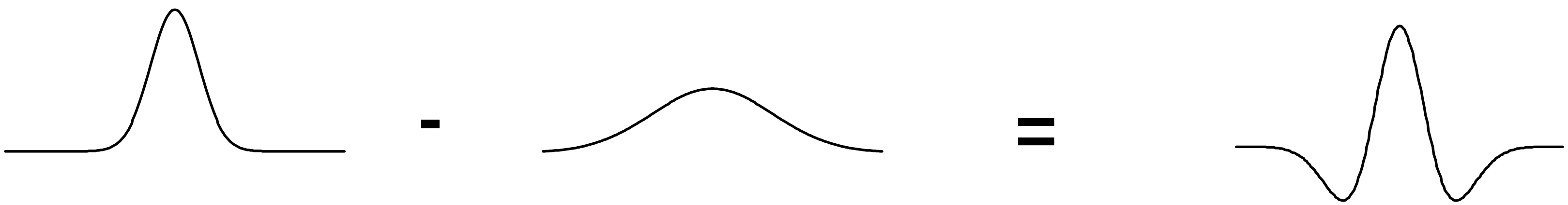
$$G_i = L_i + \mathrm{expand}(G_{i+1})$$

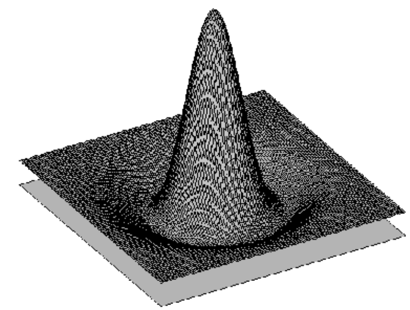


Why is this useful?

Source: Irani & Basri

Laplacian ~ Difference of Gaussian





DoG = Difference of Gaussians

Cheap approximation – no derivatives needed.

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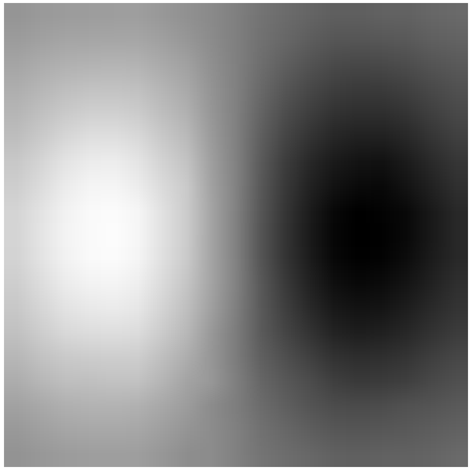
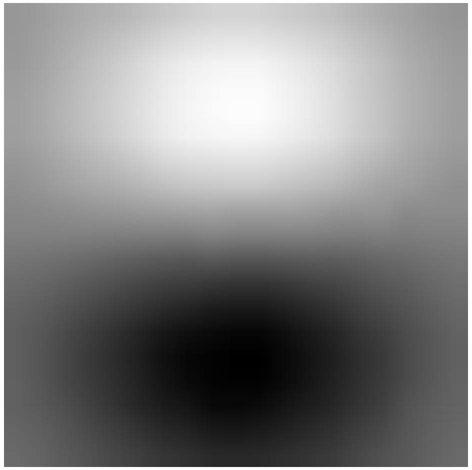
Note: Filters are Templates

- Applying a filter at some point can be seen as taking a dot-product between the image and some vector.
- Filtering the image is a set of dot products.

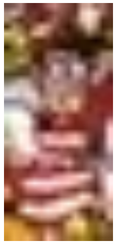
- Insight
  - Filters look like the effects they are intended to find.
  - Filters find effects they look like.





Where's Waldo?

Scene

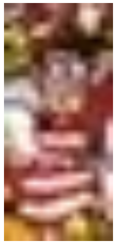


Template

Slide credit: Kristen Grauman

Where’s Waldo?

Detected template

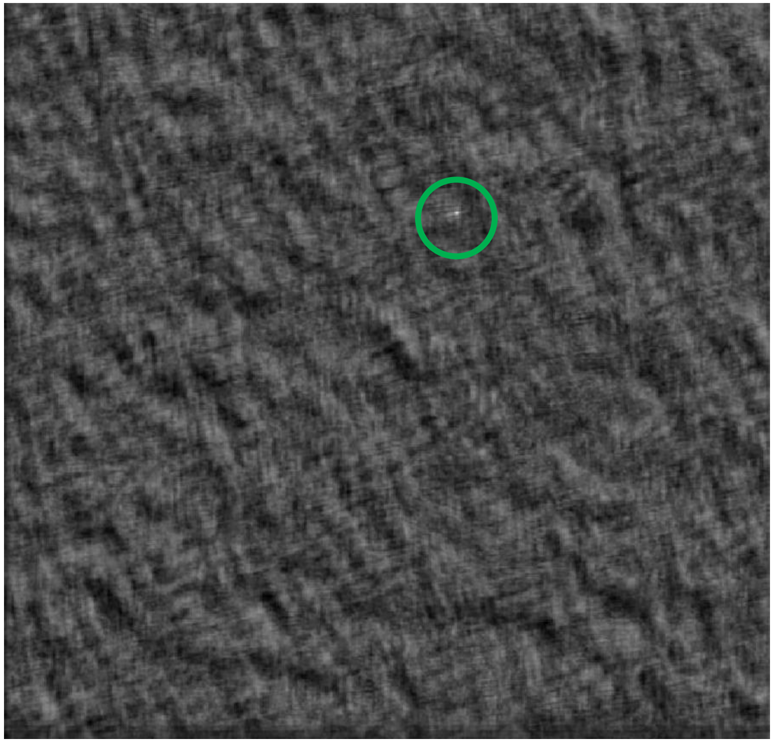


Template

Slide credit: Kristen Grauman

Where's Waldo?

Detected template



Correlation map

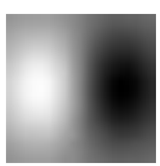
Slide credit: Kristen Grauman

Correlation as Template Matching

- Think of filters as a dot product of the filter vector with the image region
  - Now measure the angle between the vectors

$$a \cdot b = |a||b| \cos\theta \qquad \cos\theta = \frac{a \cdot b}{|a||b|}$$

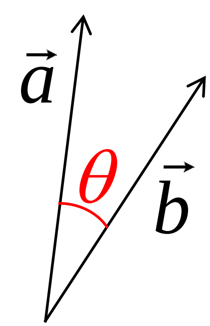
  - Angle (similarity) between vectors can be measured by normalizing the length of each vector to 1 and taking the dot product.



Template

$\cdot$

Image region



Vector interpretation

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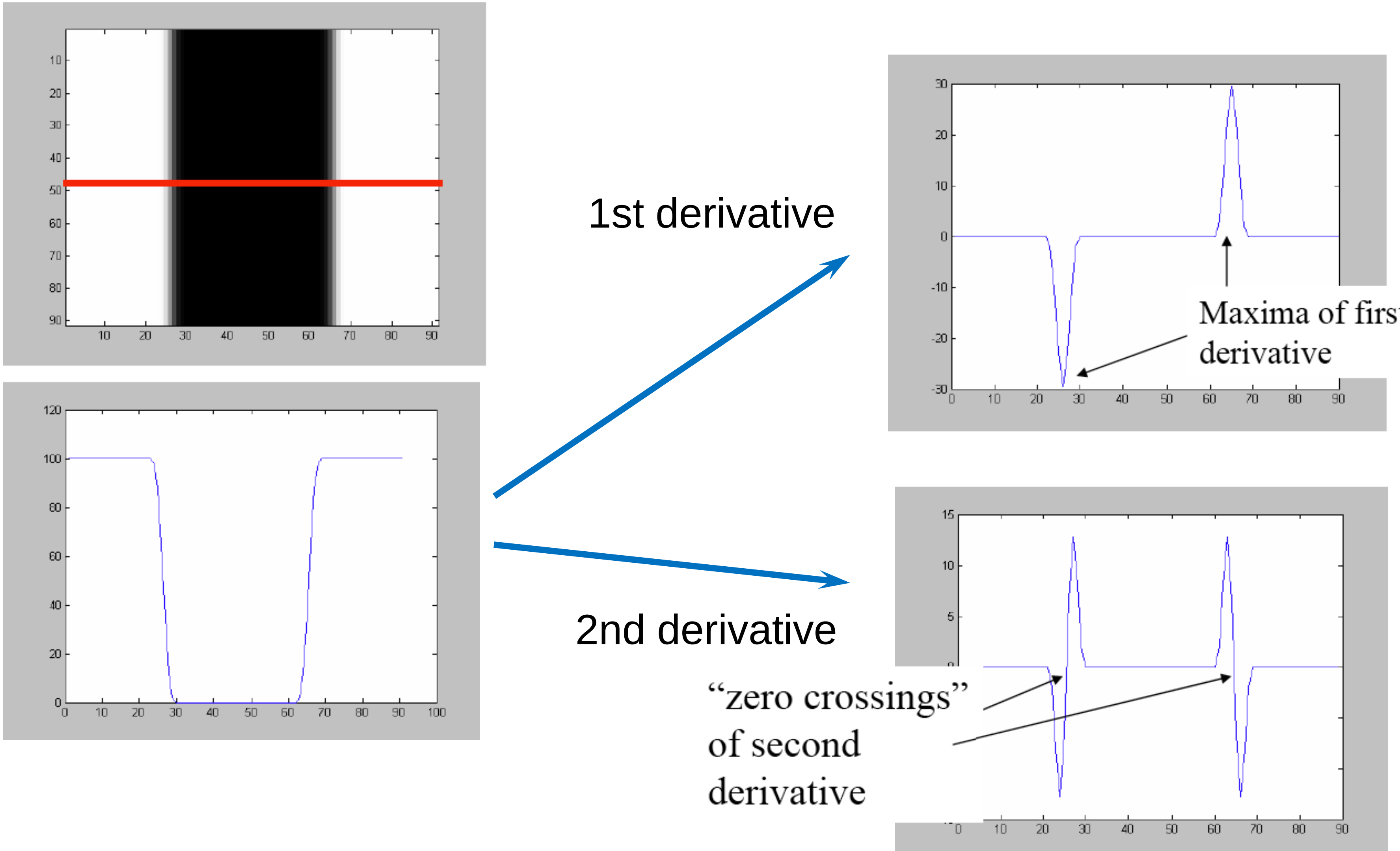
- Image gradients
  - Derivatives of Gaussian

Derivatives and Edges…



Differentiation and Convolution

- For the 2D function $f(x,y)$, the partial derivative is:

$$\frac{\partial f(x,y)}{\partial x} = \lim_{\varepsilon \to 0} \frac{f(x+\varepsilon, y) - f(x,y)}{\varepsilon}$$

- For discrete data, we can approximate this using finite differences:

$$\frac{\partial f(x,y)}{\partial x} \approx \frac{f(x+1, y) - f(x,y)}{1}$$

- To implement the above as convolution, what would be the associated filter?

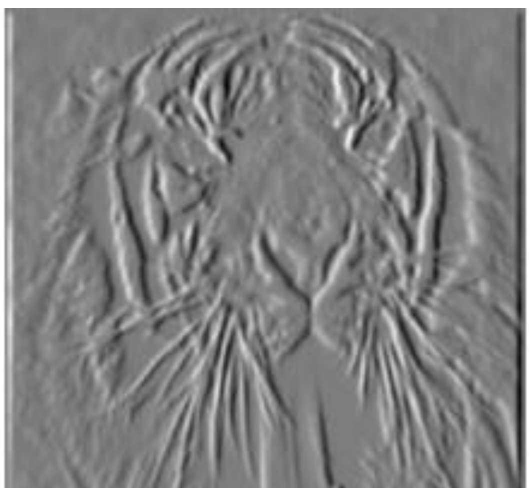
1	**-1**

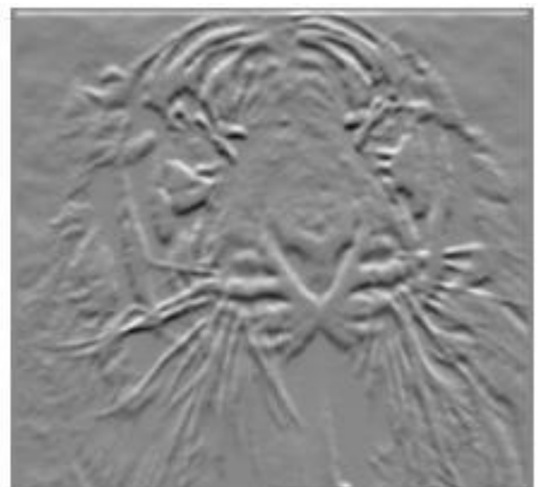
Slide credit: Kristen Grauman

Partial Derivatives of an Image

$$\frac{\partial f(x,y)}{\partial x}$$

-1	1





$$\frac{\partial f(x,y)}{\partial y}$$

-1
1

? or

1
-1

Which one shows changes with respect to x?

Slide credit: Kristen Grauman

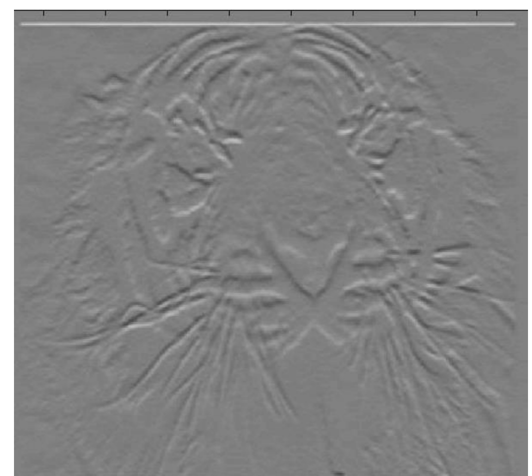
Assorted Finite Difference Filters

Prewitt: $M_x = \begin{array}{|c|c|c|}\hline -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ \hline -1 & 0 & 1 \\ \hline \end{array}$; $M_y = \begin{array}{|c|c|c|}\hline 1 & 1 & 1 \\ \hline 0 & 0 & 0 \\ \hline -1 & -1 & -1 \\ \hline \end{array}$

Sobel: $M_x = \begin{array}{|c|c|c|}\hline -1 & 0 & 1 \\ \hline -2 & 0 & 2 \\ \hline -1 & 0 & 1 \\ \hline \end{array}$; $\boxed{M_y = \begin{array}{|c|c|c|}\hline 1 & 2 & 1 \\ \hline 0 & 0 & 0 \\ \hline -1 & -2 & -1 \\ \hline \end{array}}$

Roberts: $M_x = \begin{array}{|c|c|}\hline 0 & 1 \\ \hline -1 & 0 \\ \hline \end{array}$; $M_y = \begin{array}{|c|c|}\hline 1 & 0 \\ \hline 0 & -1 \\ \hline \end{array}$

```
>> My = fspecial('sobel');
>> outim = imfilter(double(im), My);
>> imagesc(outim);
>> colormap gray;
```



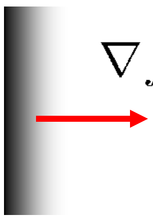
Slide credit: Kristen Grauman

Image Gradient

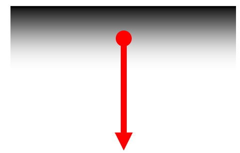
- The gradient of an image:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$$

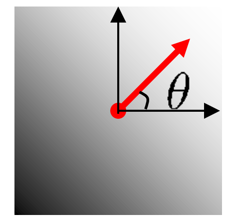
- The gradient points in the direction of most rapid intensity change



$$\nabla f = \left[\frac{\partial f}{\partial x}, 0\right]$$



$$\nabla f = \left[0, \frac{\partial f}{\partial y}\right]$$



$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$$

- The gradient direction (orientation of edge normal) is given by:

$$\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$

- The edge strength is given by the gradient magnitude

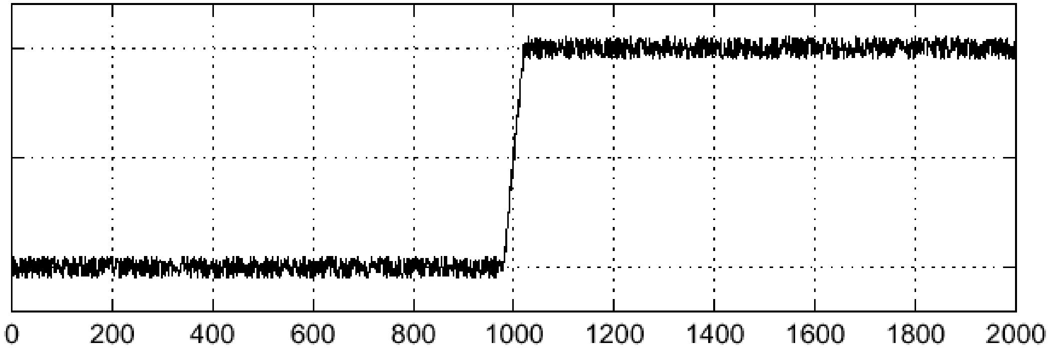
$$||\nabla f|| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

Slide credit: Steve Seitz

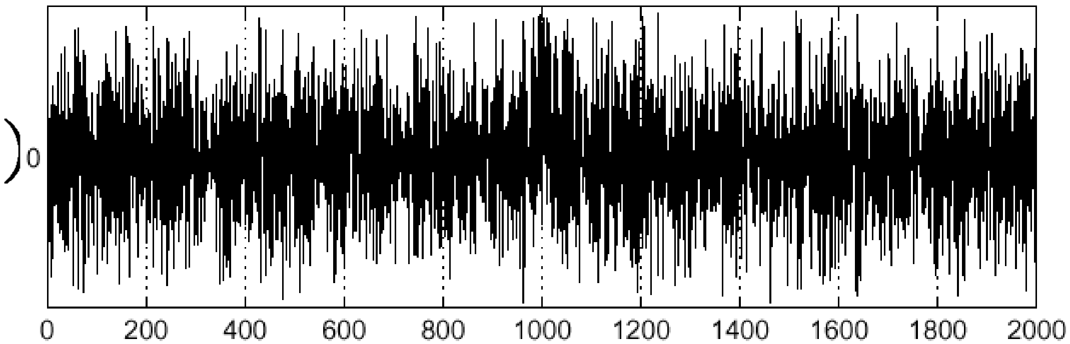
Effect of Noise

- Consider a single row or column of the image
  - Plotting intensity as a function of position gives a signal

$f(x)$



$\frac{d}{dx}f(x)$



Where is the edge?

Slide credit: Steve Seitz

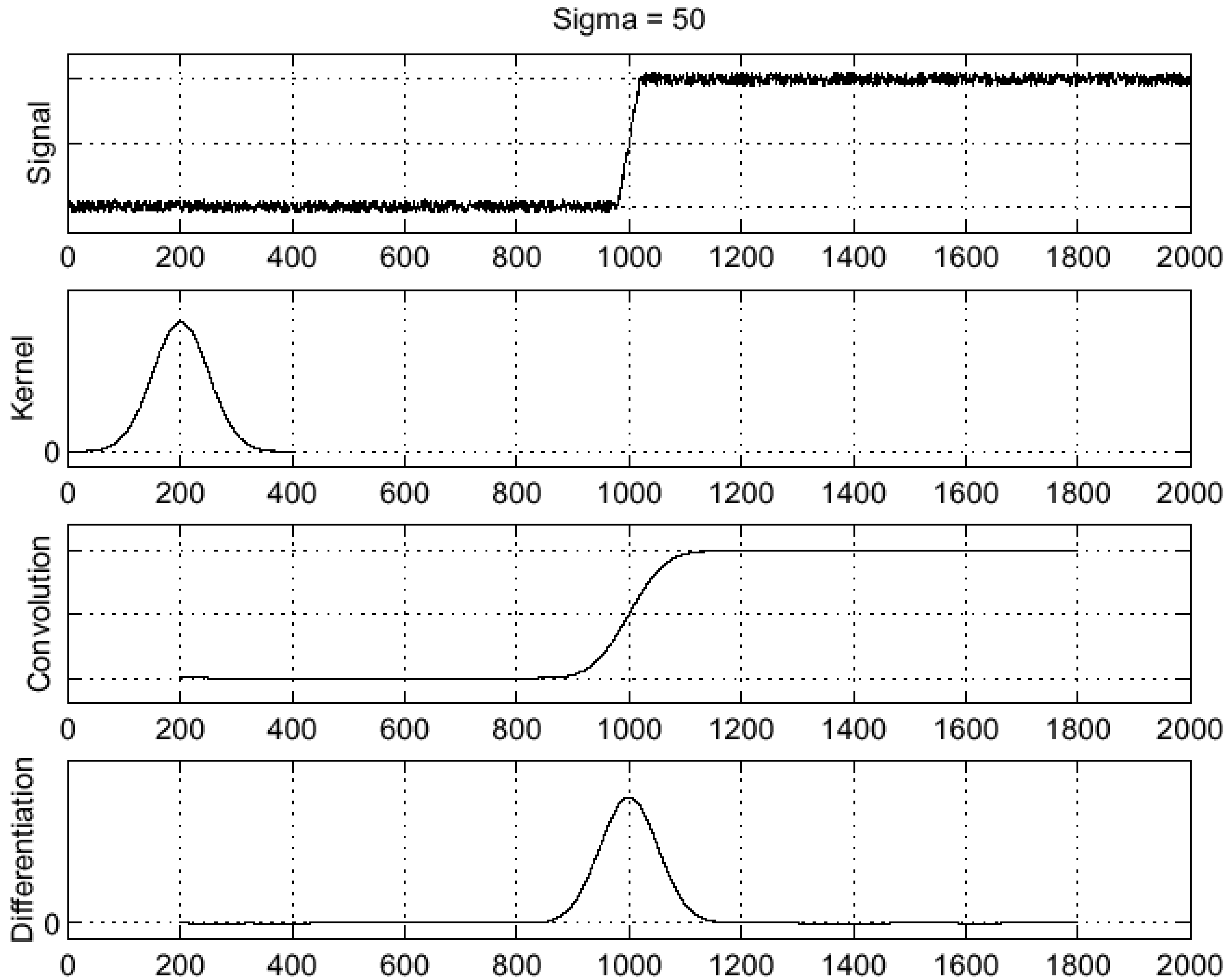
Solution: Smooth First

f

h

$h \star f$

$\frac{\partial}{\partial x}(h \star f)$



Where is the edge? Look for peaks in $\frac{\partial}{\partial x}(h \star f)$

Slide credit: Steve Seitz

Derivative Theorem of Convolution

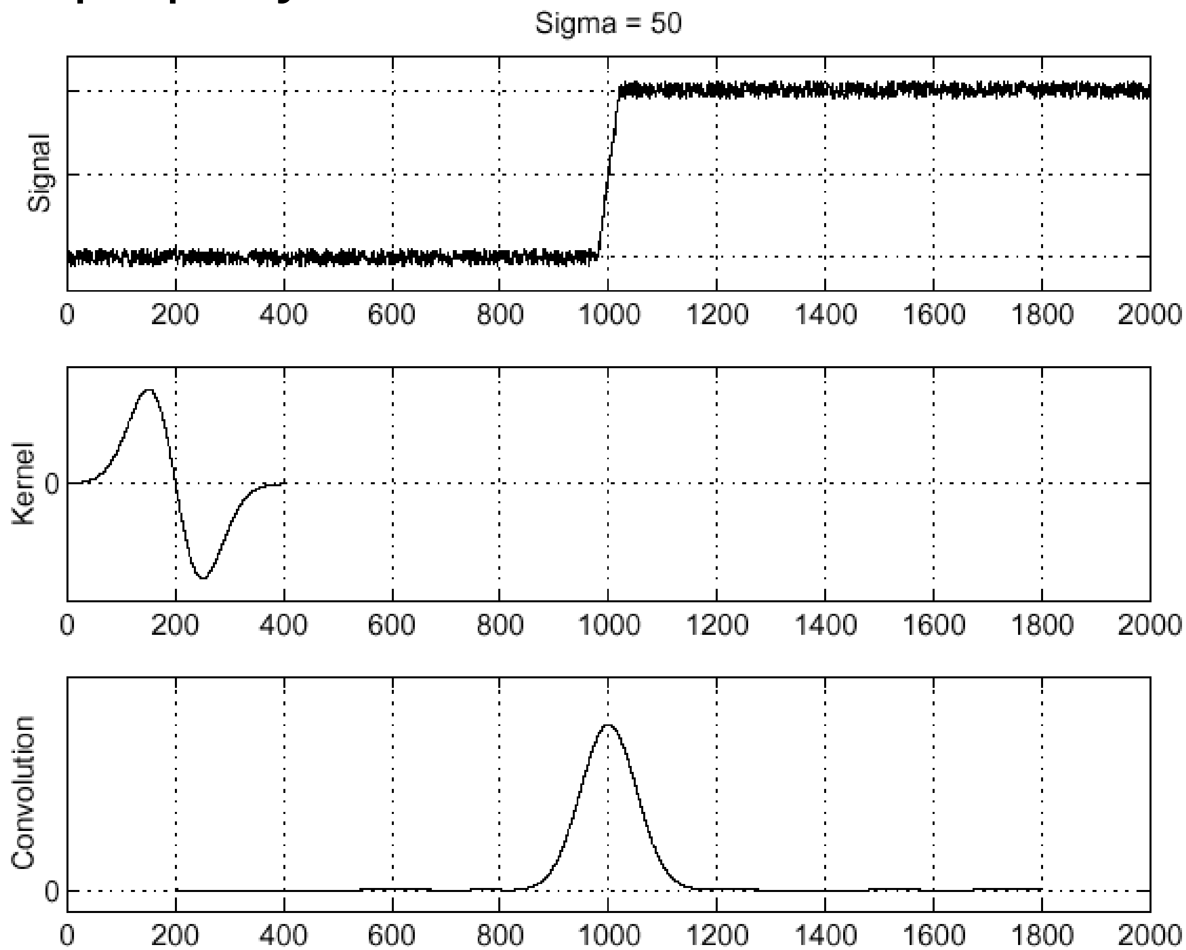
$$\frac{\partial}{\partial x}(h \star f) = (\frac{\partial}{\partial x}h) \star f$$

- Differentiation property of convolution.

f

$\frac{\partial}{\partial x}h$

$(\frac{\partial}{\partial x}h) \star f$



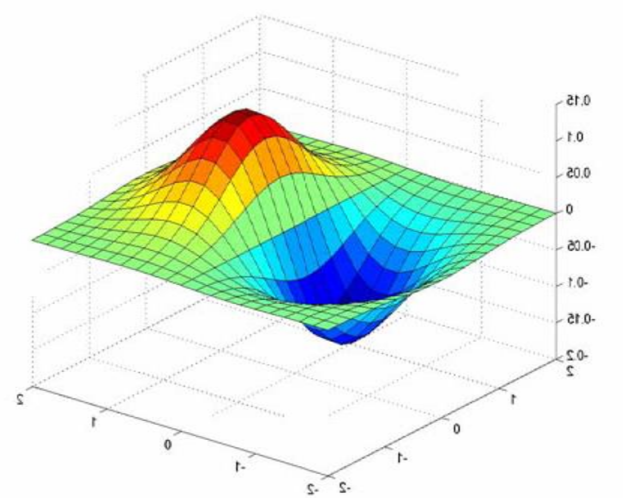
Slide credit: Steve Seitz

Derivative of Gaussian Filter

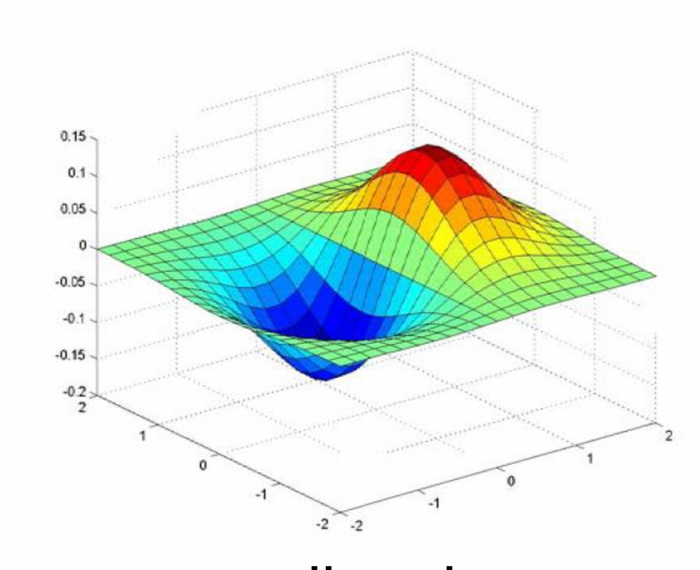
$$g * (h * I) = (g * h) * I$$

$$[1 \quad 0 \quad -1] \star \begin{bmatrix} 0.0030 & 0.0133 & 0.0219 & 0.0133 & 0.0030 \\ 0.0133 & 0.0596 & 0.0983 & 0.0596 & 0.0133 \\ 0.0219 & 0.0983 & 0.1621 & 0.0983 & 0.0219 \\ 0.0133 & 0.0596 & 0.0983 & 0.0596 & 0.0133 \\ 0.0030 & 0.0133 & 0.0219 & 0.0133 & 0.0030 \end{bmatrix}$$

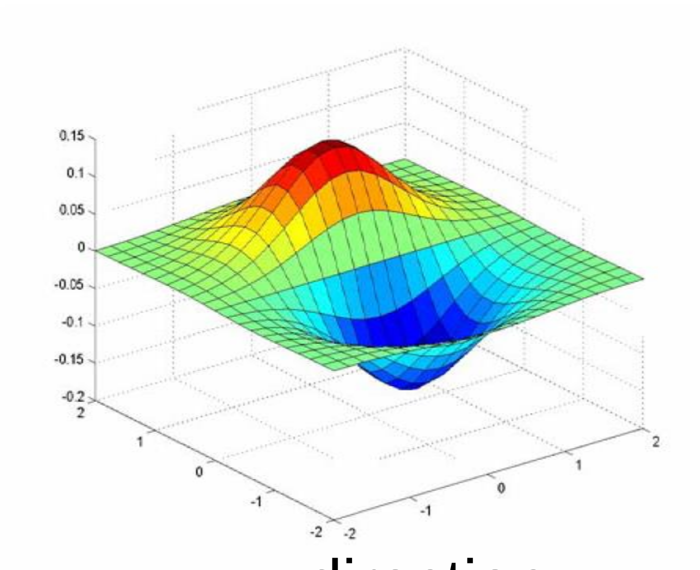
Why is this preferable?



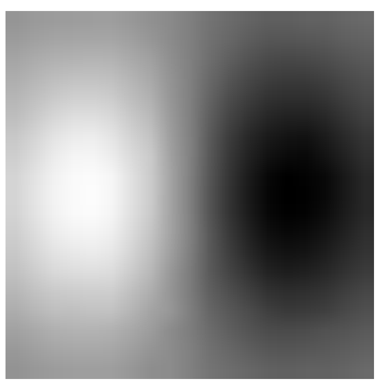
Derivative of Gaussian Filters

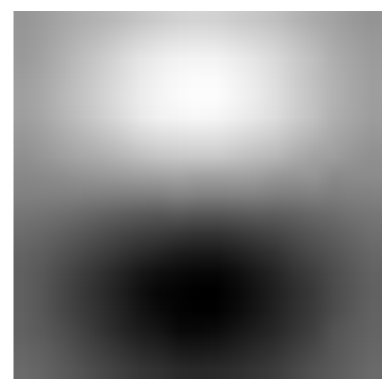


x-direction



y-direction





Source: Svetlana Lazebnik

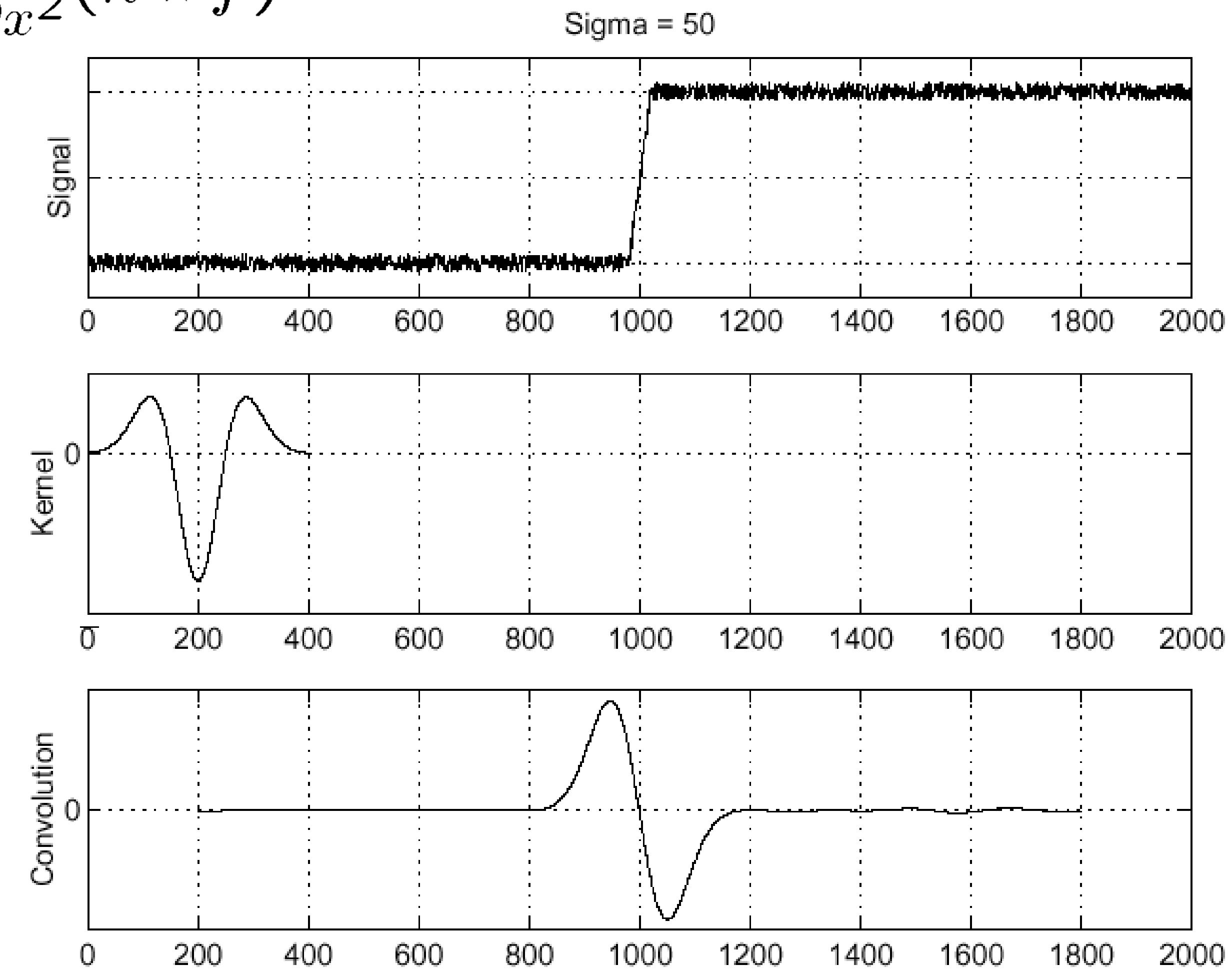
Laplacian of Gaussian (LoG)

- Consider $\frac{\partial^2}{\partial x^2}(h \star f)$

$$f$$

$$\frac{\partial^2}{\partial x^2}h$$

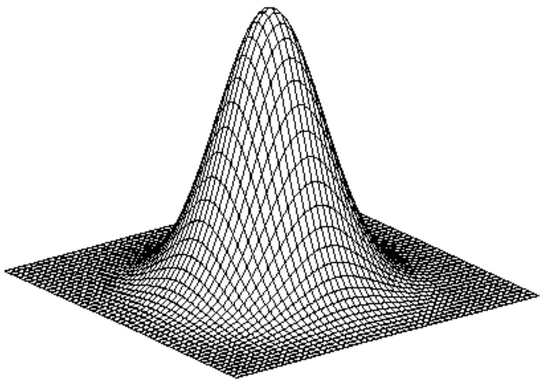
$$(\frac{\partial^2}{\partial x^2}h) \star f$$



Where is the edge? Zero-crossings of bottom graph

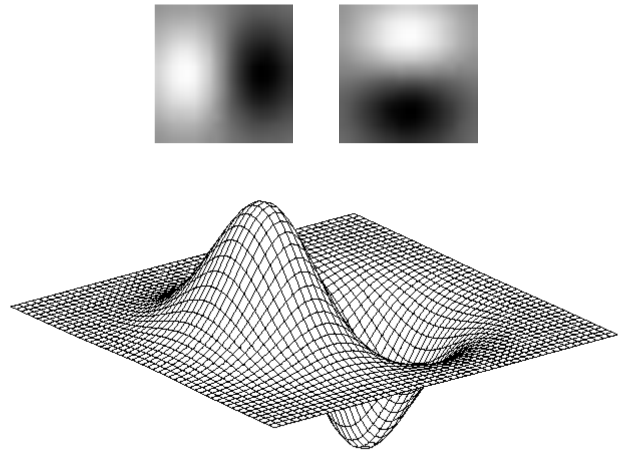
Slide credit: Steve Seitz

Summary: 2D Edge Detection Filters



Gaussian

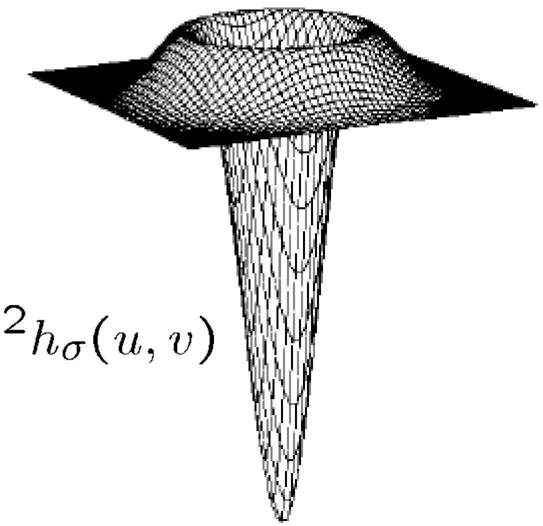
$$h_\sigma(u,v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{2\sigma^2}}$$



Derivative of Gaussian

$$\frac{\partial}{\partial x} h_\sigma(u,v)$$

Laplacian of Gaussian



$$\nabla^2 h_\sigma(u,v)$$

- ∇^2 is the Laplacian operator:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Slide credit: Kristen Grauman

References and Further Reading

- Background information on linear filters and image gradients can be found in Chapter 3 of the Szeliski book.

R. Szeliski
Computer Vision – Algorithms and Applications
Springer, 2010
(draft version available online)